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United States Patent Application Publication	20250276384
Kind Code	A1
Publication Date	September 04, 2025
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MASS AND HEAT FLOW IN ADDITIVE MANUFACTURING SYSTEMS WITH MACHINE LEARNING CONTROL

Abstract

An additive manufacturing system may include an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a plurality of mass sensors, each mass sensor associated with a portion of the additive manufacturing system; a plurality of heat sensors; and one or more computing devices. The computing device(s) are configured to receive data from the plurality of mass sensors; determine an overall mass flux based on the data from the mass sensors; control the powder delivery device based on the overall mass flux; receive data from the plurality of heat sensors; determine an overall heat flux based on the data from the heat sensors; and control the energy delivery device based on the overall heat flux.

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Family ID: **96881600**

Appl. No.: **18/593575**

Filed: **March 01, 2024**

Publication Classification

Int. Cl.: **B22F12/90** (20210101); **B22F10/28** (20210101); **B22F10/34** (20210101); **B22F10/36** (20210101); **B22F10/85** (20210101); **B22F12/40** (20210101); **B22F12/53** (20210101); **B33Y10/00** (20150101); **B33Y30/00** (20150101); **B33Y50/02** (20150101)

U.S. Cl.:

CPC **B22F12/90** (20210101); **B22F10/28** (20210101); **B22F10/34** (20210101); **B22F10/36** (20210101); **B22F10/85** (20210101); **B22F12/40** (20210101); **B22F12/53** (20210101); **B33Y10/00** (20141201); **B33Y30/00** (20141201); **B33Y50/02** (20141201); B22F2203/00 (20130101); B22F2998/10 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures.

SUMMARY

[0003] As an example, an additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a plurality of sensors, each mass sensor associated with a portion of the additive manufacturing system; a plurality of heat sensors; and one or more computing devices configured to: receive data from the plurality of mass sensors; determine an overall mass flux based on the data from the plurality of mass sensors; receive data from the plurality of heat sensors; determine an overall heat flux based on the data from the plurality of heat sensors; and input, into one or more machine learning models, the overall mass flux and the overall heat flux; and control, based at least in part on outputs from the one or more machine learning models, the powder delivery device and the energy delivery device.

[0004] As another example, a method includes receiving, by one or more computing devices, data from a plurality of mass sensors of an additive manufacturing system, wherein the additive manufacturing system comprises an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of a component, a powder delivery device configured to direct a powder stream toward the melt pool, the plurality of mass sensors, each mass sensor associated with a portion of the additive manufacturing system, and a plurality of heat sensors; determining, by the one or more computing devices, a mass flux based on the data from the plurality of mass sensors; receiving, by the one or more computing devices, data from the plurality of heat sensors; determining, by the one or more computing devices, a heat flux based on the data from the plurality of heat sensors; inputting, by the one or more computing devices and into one or more machine learning models, the overall mass flux and the overall heat flux; and controlling, based at least in part on outputs from the one or more machine learning models, the powder delivery device and the energy delivery device.

[0005] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. **1** is a conceptual block diagram illustrating mass flow monitoring aspects of an example additive manufacturing system that includes a powder source mass sensor, a powder flow monitoring system, and a topology sensor configured to monitor mass flow of powder within the additive manufacturing system during an additive manufacturing technique.

[0007] FIG. **2** is a conceptual and schematic diagram illustrating an example powder flow monitoring system configured to monitor powder flow between a powder delivery device and a build surface during the additive manufacturing technique.

[0008] FIG. **3** is a conceptual diagram illustrating an example of portions of a powder stream imaged by a powder flow monitoring system.

[0009] FIG. **4** is an example calibration curve of particle detections versus mass flow.

[0010] FIG. **5** is a conceptual block diagram illustrating heat flow monitoring aspects of the example additive manufacturing system of FIG. **1**, including an optical system for observing thermal emissions around a melt pool and a thermal camera for monitoring a size of the melt pool.

[0011] FIG. **6** is a conceptual block diagram illustrating an example optical system for observing thermal emissions around a melt pool formed during the additive manufacturing technique.

[0012] FIG. **7** is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique.

[0013] FIG. **8** is a flowchart illustrating an example technique for fabricating a component with closed loop control.

[0014] FIG. **9**A is a conceptual diagram illustrating an example machine learning model.

[0015] FIG. **9**B is a conceptual diagram illustrating an example training process for a machine learning model.

DETAILED DESCRIPTION

[0016] The disclosure generally describes techniques and systems for monitoring mass flux and heat flux in a blown powder additive manufacturing technique, such as a directed energy deposition (DED) technique. During blown powder additive manufacturing, a component is built up by adding material to the component in sequential layers. The final component is composed of a plurality of layers of material. In some blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool. A powder delivery device may deliver a powder to the melt pool, where at least some of the powder at least partially melts and is joined to the melt pool and, thus, substrate.

[0017] The properties of the final component, including the presence or absence of material defects and the resulting microstructure, are a function of a number of variables related to mass flux and heat flux. As such, measurement of mass flux and heat flux within the blown powder additive manufacturing system may enable characterization or prediction of final component properties, control of the blown powder additive manufacturing technique, quality assurance for the final component, development of new blown powder additive manufacturing techniques, and the like.

[0018] An additive manufacturing system may include a plurality of sensors for sensing mass flow at various points along the powder flow and a plurality of sensors for sensing heat flow within the system. For instance, the mass flow sensors may include a mass sensor associated with a powder source, a powder flow monitoring system sensing powder flow between an output of a powder delivery device and the melt pool, and a topology sensor for measuring a topology of material added to the melt pool. The thermal sensors may include at least one sensor for monitoring a size and/or temperature of the melt pool and at least one sensor for monitoring a heat flow (e.g., cooling rate) around the melt pool. The sensors may output data to a computing device, which analyzes the received sensor data. By monitoring mass flow at different points along the powder flow and monitoring heat flow in multiple ways, the system described herein may enable a more complete understanding of mass and heat flux within the system.

[0019] Understand the mass and heat flux within the system may give an operator insight into a build in process. The operator may adjust operating parameters (e.g., key process variables (KPVs)) based on the mass flux and/or the heat flux. However, an adage of machine control is to "only turn one knob at a time." Put differently, it may be undesirable for the operator or other controller to adjust (e.g., alter, modify, etc.) multiple of the operating parameters at once. In particular, adjusting multiple of the operating parameters at the same time may yield unpredictable results. However, only adjusting one of the operating parameters at a time may extend the amount of time it takes the operator or controller to "react" to changes in the build.

[0020] In accordance with one or more aspects of this disclosure, an additive manufacturing system may include a controller that executes one or more machine learning models to adjust two or more of the operating parameters in parallel. For instance, the controller may input the determined heat flux and the mass flux into the one or more machine learning models and adjust, based on output of the one or more machine learning models, the two or more of the operating parameters in parallel. In some examples, the two or more operating parameters adjusted by the controller may be non-linearly related to each other. With such a non-linear relationship, it may not be feasible for an operator or traditional controllers to adjust the two or more operating parameters in parallel. However, with the use of the machine learning models, such parallel adjustment of multiple non-linearly related operating parameters may be accomplished. By adjusting two or more operating parameters in parallel, the controller may more quickly "react" to changes in the build.

[0021] FIG. **1** is a conceptual block diagram illustrating mass flow monitoring aspects of an example additive manufacturing system **10** that includes a powder source mass sensor **44**, a powder flow monitoring system **18**, and a topology sensor **48** configured to monitor mass flow of powder within additive manufacturing system **10** during an additive manufacturing technique. In the example illustrated in FIG. **1**, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, PFMS **18**, a stage **20**, a powder source **42**, powder source mass sensor **44**, and topology sensor **48**. Computing device **12** is operably connected to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and topology sensor **48**. FIG. **1** thus illustrates mass flow monitoring aspects of example additive manufacturing system **10**. To simplify illustration of FIG. **1** and improve clarity of the figure, heat flow monitoring aspects of additive manufacturing system **10** are shown in FIG. **5** and described below with reference to FIG. **5**.

[0022] In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. For example, stage **20** may be translatable and/or rotatable along at least one axis to position component **22** relative to energy delivery device **16** and/or powder delivery device **14**. Similarly, energy delivery device **16** and/or powder delivery device **14** may be translatable and/or rotatable along at least one axis to position energy delivery device **16** and/or powder delivery device **14**, respectively, relative to component **22**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0023] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream.

[0024] System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0025] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).
[0026] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder. The combined powder defines powder stream **30**. In some examples, powder delivery device **14** includes a single nozzle, which may be point nozzle, or a single nozzle that is an annular channel. In other examples, powder delivery device **14** includes a plurality of nozzles (e.g., three nozzles or four nozzles). Regardless of the number of nozzles, powder delivery device **14** may output a powder stream that is focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. **1** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**.
[0027] At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**. At least some of the powder that impacts melt pool **32** may be joined to component **22**.
[0028] In some examples, powder delivery device **14** may be mechanically coupled or attached to energy delivery device **16** to facilitate delivery of powder stream **30** and energy **34** for forming melt pool **32** to substantially the same location adjacent to component **22**.
[0029] Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique.
[0030] In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**.
[0031] In some examples, at least a portion of energy delivery device **16** and powder delivery device **14** may be combined or attached to each other. For example, a deposition head (e.g., deposition head **54** of FIG. **2**) may include part of powder delivery device **14** (e.g., internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**) and part of energy delivery device **16** (e.g., the energy delivery head). As shown in FIG. **1**, in some examples, energy delivery device **16** may be arranged of configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) **56** for forming powder stream **30** and directing powder stream **30** toward build surface **28**.
[0032] System **10** also includes powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between

powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light source. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**.

[0033] The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz. Because of the velocity of the powder in powder stream **30**, even such a frame rate may image only a fraction of the powder flowing between powder delivery device **14** and build surface **28**.

[0034] In some examples, PFMS **18** also includes a housing configured to enclose the illumination device and the imaging device. The housing may be configured to protect the illumination device and the imaging device from damage due to the harsh conditions to which PFMS **18** may be exposed during use. For example, the housing may protect the illumination device and the imaging device from powder deflections from powder stream **30** off build surface **28**, may cool the illumination device and the imaging device to remove heat incident on PFMS **18** from melt pool **32** and energy delivery device **16**, or the like.

[0035] PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16**, e.g., in the x-y plane shown in FIG. **1**. This may help maintain a relative x-y position of PFMS **18** and the image plane of the imaging device relative to powder stream **30**. This may facilitate analysis of image data captured by the imaging device.

[0036] PFMS **18** may be movable in the z-axis direction of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may enable movement of image plane **38** along the z-axis of FIG. **1** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may allow PFMS **18** to image powder stream **30** at different positions between powder delivery device **14** and build surface **28**. In this way, PFMS **18** may analyze powder stream **30** along powder stream **30** to help determine parameters of powder stream **30** along its length.

[0037] In some example, PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16** and movable parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28** by an adjustable z-stage **40**. Adjustable z-stage **40** may be attached to energy delivery device **16**, powder delivery device **14**, or a portion of system **10** that moves energy delivery device **16** and/or powder delivery device **14**, such that PFMS **18** moves in the x-y axis in registration with energy delivery device **16** and/or powder delivery device **14**.

[0038] Adjustable z-stage **40** may be controlled by computing device **12** to position PFMS **18** and image plane **38** relative to powder stream **30**. Further, computing device **12** may control adjustable z-stage **40** to move PFMS **18** vertically and out of the way to allow powder delivery device **16** and energy delivery device **16** access to physically constrained areas, e.g., between vanes of a doublet or triplet of a nozzle guide vane for a gas turbine engine.

[0039] System **10** further includes a topology sensor **48**. Topology sensor **48** is configured to monitor an amount of powder captured by melt pool **32** by imaging melt pool **32** and the added material, allowing the mass to be quantified (e.g., by computing device **12**) using the dimensions of the added material and density of the material (powder). In some examples, topology sensor **48** includes a laser and a sensor (e.g., an imaging device), which senses laser light reflected by the structure being imaged (e.g., melt pool **32** and the added material). The laser may have a defined wavelength, which may affect the resolution of the topology sensor **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology sensor **48** is a great as about 10 microns (e.g., about 6 microns).

[0040] In some examples, topology sensor **48** may be positioned substantially directly above component **22** and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. In other examples, topology sensor **48** may be positioned at an offset with respect to component **22** such that the sensor senses depth information without using an interferometer.

[0041] In some examples, topology sensor **48** may be integral with system **10**, e.g., disposed within the enclosure or working area of system **10**. In other examples, topology sensor **48** may be an add-on component to system **10**. For example, the enclosure in which the additive manufacturing technique is performed may include a transparent window, and topology sensor **48** may be positioned outside of the enclosure and may image component **22** through the transparent window.

[0042] Although a topology sensor **48** is described in the examples of this disclosure, in other examples, another metrology device may be utilized to determine the amount of powder captured by melt pool **32**. For example, another type of light source may be used. In some examples, if another type of light source is used, component **22** or stage **20** may include one or more features that serve as indicators of scale.

[0043] Computing device **12** is configured to control components of system **10** and may include, for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** is configured to control operation of system **10**, including, for example, powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48**. Computing device **12** may be communicatively coupled to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** using respective communication connections. As shown in FIG. **1**, computing device **12** may include machine learning model (ML model) **200**, which may represent one or more ML models. In some examples, the communication connections may include network links, such as Ethernet, ATM, or other network connections. Such connections may be wireless and/or wired connections. In other examples, the communication connections may include other types of device connections, such as USB, IEEE 1394, or the like.

[0044] Although FIG. **1** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**. In general, control and processing functions described herein may be divided among one or more computing devices. For instance, system **10** may include controller for energy delivery device **16**, powder delivery device **14**, and stage **20**, a separate controller for PFMS **18**, and a separate computing device for analyzing data obtained by PFMS **18**, mass sensor **44**, and topology sensor **48**. As another example, system may include a dedicated controller for each of energy delivery device **16**, powder delivery device **14**, stage **20**, PFMS **18**, and topology sensor **48**, and a separate computing device for coordinating control of powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** and analyzing data obtained by PFMS **18** powder source mass sensor **44**, and/or topology sensor **48**. Other examples of computing system architectures for controlling system **10** and analyzing data obtained from system **10** will be apparent and are within the scope of this disclosure.

[0045] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, stage **20**, and/or topology sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40** and/or topology sensor to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**.

[0046] Computing device **12** may be configured to control system **10** to deposit layers **24** and **26** to form component **22**. As shown in FIG. **1**, component **22** may include a first layer **24** and a second layer **26**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including first layer **24** and second layer **26**, the technique may be extended to components **22** with more complex geometry and any number of layers.

[0047] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** to form, on a surface **28** of first layer of material **24**, a second layer of material **26** using an additive manufacturing technique. Computing device **112** may control energy delivery device **16** and powder delivery device **14** based on one or more operating parameters. In general, the operating parameters may be KPVs that impact the build, such as laser power, travel speed, pause time, dwell time, build strategy, power feed rate, layer thickness, spot size/power density, working distance, etc. Further details of example operating parameters are discussed below. Computing device **12** may control energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**. Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** then may control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26** is complete. Computing device **12** may control powder delivery device **14** and energy delivery device **16** similarly until all layers are formed to define a completed component **22**.

[0048] In accordance with one or more aspects of this disclosure, computing device **12** may utilize ML model **200** to control two or more operating parameters of the plurality of operating parameters in parallel. For instance, computing device **12** may input, to ML model **200**, variables determined based on outputs of one or more sensors (e.g., PFMS **18**, mass sensor **44**, and/or topology sensor **48**). ML model **200** may generate outputs based on said inputs. The outputs provided by ML model **200** may include adjusted operating parameters. Computing device **12** may control energy delivery device **16** and powder delivery device **14** based on the updated operating parameters.

[0049] FIG. **2** is a conceptual and schematic diagram illustrating an example powder flow monitoring system **50** configured to monitor powder flow between a powder delivery device **52** and a build surface (not shown in FIG. **2**) during an additive manufacturing technique. Powder delivery device **52** may be an example of powder delivery device **14** of FIG. **1**, and PFMS may be an example of PFMS **18** of FIG. **1**. Similar to powder delivery device **14**, powder delivery device **52** may be controlled using one or more machine learning models.

[0050] Powder delivery device **52** includes a deposition head **54** that carries a plurality of powder nozzles **56**. Plurality of powder nozzles **56** output a powder stream **58** toward the build surface. As shown in FIG. **2**, the powder stream **58** may be focused at a focal plane, such that powder stream **58** is converging toward the focal plane and diverging away from the focal plane.

[0051] PFMS **18** includes a housing **60** (also referred to as an enclosure), which encloses an imaging device **62** and an illumination device **64**. In some examples, imaging device **62** may be a high-speed camera and illumination device **64** may be laser illuminator. Housing **60** is attached to an adjustable z-stage **66** by a bracket **68**.

[0052] Housing **60** is configured to enclose imaging device **62** and illumination device **64** and help protect imaging device **62** and illumination device **64** from a surrounding environment. For instance, housing **60** may be configured to surround imaging device **62** and illumination device **64** and prevent any powder that reflects from the build surface toward PFMS **18** from impacting imaging device **62** or illumination device **64**.

[0053] Further, housing **60** may be configured to cool imaging device **62** and illumination device **64**. Imaging device **62** and illumination device **64** may be exposed to heat from the melt pool at the build surface and energy from the energy delivery device. Imaging device **62** and illumination device **64** may be relatively sensitive to heat and have improved operational lifetime if maintained and operated below a certain temperature. PFMS **50** may include a cooling system **70** configured to remove heat from within housing **60** to cooling imaging device **62** and illumination device **64**. For instance, cooling system **70** may include cooling fluid circuit through which a cooling fluid flows, and housing **60** may include part of the cooling circuit. In some examples, housing **60** may be formed from a material having relatively high thermal conductivity, such as aluminum, to help transfer heat from within housing **60** to cooling system **70** (e.g., a cooling fluid flowing through cooling system **70**).

[0054] As described above, PFMS **50** may be configured to measure powder flow of powder stream **58** (FIG. **2**) at one or more axial (or longitudinal) locations of powder stream **58** and determine one or more parameters associated with the powder flow. For instance, illumination device **64** may illuminate powder of powder stream **58** in a plane oriented substantially orthogonal to a longitudinal axis that extends from powder delivery device **52** to the build surface. PFMS **50** may be positioned at a selected axial or longitudinal location to image a selected axial or longitudinal position between powder delivery device **52** and the build surface. Imaging device **62** may be configured to image at least some of the illuminated powder. FIG. **3** is a conceptual diagram illustrating an example of portions of a powder stream imaged by a powder flow monitoring system.

[0055] As shown in FIG. **3**, since powder is flowing in powder stream **58** at a relatively high velocity, imaging device **62** may not capture images of all the powder in powder stream **58**. The fraction of powder that imaging device **62** captures images of may be a function of average powder velocity at the image plane and a frame rate or capture speed of imaging device **62**. This is represented in FIG. **3** as “sampled” particles and “missed population” particles. The fraction of particles imaged by imaging device **62** may, in some examples, be less than about 50%, less than about 40%, less than about 30%, less than about 25%, less than about 20%, or less than about 15%.

[0056] PFMS **50** may include a computing device (e.g., computing device **12** of FIG. **1**) configured to analyze images captured by imaging device **62** to identify a number of particle detections in each captured image and, optionally, derive further parameters from the number of particle detections. As such, computing device **12** may be configured to receive image data representing an image captured by imaging device **62**. The image data may include representations of illuminated powder of powder stream **58**, as imaged by imaging device **62** (e.g., as captured in an image frame by imaging device **62**). Computing device **12** may be configured to generate a representation of powder stream based on the image data and output the representation of the powder stream for display at a display device.

[0057] For instance, computing device **12** may be configured determine a powder mass flow represented by the image data. To do so, computing device **12** may be configured to identify a number of powder particles within each image frame. In some examples, computing device **12** additionally may be configured to identify a size and/or shape of each powder particle within each image frame. Computing device **12** may be configured to implement any suitable image analysis technique to identify powder particles, and, optionally, size and/or shape of powder particles.

[0058] Once computing device **12** has identified a number of powder particles within an image frame, computing device **12** may be configured to determine a mass flow based on the number of powder particles. For example, computing device **12** may be configured to determine the mass flow based on a calibration equation or calibration curve. FIG. **5** is an example calibration curve of particle detections versus mass flow. As shown in FIG. **4**, the relationship between particle detections may be substantially linear.
[0059] The relationship between particle detections and mass flow may be determined experimentally. For instance, the relationship between particle detections and mass flow may be determined for each powder type (e.g., composition, size distribution, or both), as each powder type may have a different relationship between particle detections and mass flow. The relationship may be determined experimentally by flowing a known mass of powder at a known rate, and imaging the powder. By doing this multiple times at multiple rates, the calibration curve may be generated. The curve, in the form of an equation, a look-up table, or the like, may be stored in computing device **12**, and computing device **12** may use the calibration curve to determine mass flow of a similar type of powder at a different flow rate based on particle detections.
[0060] In some examples, computing device **12** may receive image data representative of a sequence of images of illuminated powder in powder stream **58**. Each image may be associated with a time. As such, computing device **12** may select one or more images of the sequence of images and analyze the one or more images. For each selected image, computing device **12** may be configured to identify a number of particle detections and, optionally, determine a mass flow associated with powder stream **58** for each image frame.
[0061] As described above, system **10** may include both mass flow monitoring and heat flow monitoring, but FIG. **1** illustrates only mass flow monitoring aspects of system **10**. FIG. **5** is a conceptual block diagram illustrating heat flow monitoring aspects of the example additive manufacturing system **10** of FIG. **1**, including an optical system **80** for observing thermal emissions around melt pool **32** and a thermal camera **82** (e.g., pool monitor) for monitoring a size and/or temperature of melt pool **32**. Identical reference numerals in FIGS. **1** and **5** refer to the same parts. Further, those common parts are the same or substantially identical, aside from any differences described herein.
[0062] As shown in FIG. **5**, system **10** includes an optical system **80**. During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy **34** at first layer **24** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **24**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **24** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **24**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **24** affect the microstructure of the component **22** formed using the additive manufacturing technique.
[0063] In many cases, energy **34** output by energy delivery device **16** is very high temperature and the intensity of its thermal emissions is significantly greater than the intensity of thermal emissions from melt pool **32** and the surrounding areas. Similarly, thermal emissions intensity at and near the center of melt pool **32** may be significantly greater than the intensity of thermal emissions near the edge of melt pool **32** and in areas surrounding melt pool **32**. Because of this, it may be difficult to accurately measure temperature and cooling rate of areas near the edge of melt pool **32** and in areas surrounding melt pool **32**. This results in difficulty predicting and controlling microstructure of the additively manufactured component **22**.
[0064] Optical system **80** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **80** may include a visible light imaging device, an infrared imaging device,

or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range.

[0065] The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device. In some examples, at least a portion of the optical train is coaxial with the axis at which energy delivery device **16** outputs energy, and the at least a portion of the optical train may be attached to or otherwise configured to move with the portion of energy delivery device **16** that directs or focuses energy **34** at or near the surface of component **22**. In this way, optical system **80** may move with energy delivery device **16** and track melt pool **32** as melt pool **32** moves across component **22**, without needing to correct for any offsets between energy delivery device **16** and optical system **80** and/or needing to correct for geometry of component **22**. In other examples, the optical train may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** may be configured to compensate for the offset and any affects this may have on the imaging, including shadowing, interference, geometry of component **22**, or the like.

[0066] Optical system **80** may include an occulting device. The occulting device is configured to reduce or block emissions (e.g., thermal emissions) that originate from the energy output by energy delivery device **16** and/or near a center of melt pool **32**, which otherwise obfuscate emissions from solidifying regions of material at or near the edge of melt pool **32** and outside of melt pool **32**. The occulting device may be a rigid occulting device or a dynamic occulting device. A rigid occulting device reduces or blocks emissions from a fixed region, e.g., from the energy **34** output by energy delivery device **16**. For instance, a rigid occulting device may include a device with fixed dimensions that is opaque to wavelengths of interest. As another example, a rigid occulting device may include an apodizing lens in which a center of the lens if substantially opaque to wavelengths of interest and opacity decreases as a function of radius.

[0067] A dynamic occulting device is configured to be controlled to occult different regions, e.g., different sizes and/or shapes. A dynamic occulting device may include a rigid occulting device that is mounted to a device that can translate the rigid occulting device along and/or perpendicularly to the optical axis. As another example, a dynamic occulting device may include an opaque and viscous liquid, such as mercury, contained between two substrates. The substrates are substantially transparent to the wavelength(s) of interest. One or both of the substrates may be movable relative to the other substrate to control the distance between the substrates. By reducing the distance between the substrates, the size of the occulting region may increase. By increasing the distance between the substrates, the size of the occulting region may decrease. As a third example, a dynamic occulting device may include a digital micromirror device. Computing device **12** may be configured to control the micromirrors of the digital micromirror device to direct emissions that originate from energy **34** output by energy delivery device **16** and/or near a center of the melt pool away from the imaging device. A digital micromirror device may enable control of both the size and shape of the region of emissions that are occulted.

[0068] FIG. **6** is a conceptual block diagram illustrating an example optical system **80** for observing thermal emissions at and/or around a melt pool **32** formed during an additive manufacturing technique. Optical system **80** includes an optical train that includes first imaging optics **92**, occulting device **94**, second imaging optics **96**, and imaging device **98**. Imaging device **98** may be any suitable imaging device, including, for example, a visible light imaging device, an infrared imaging device, an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range, a two color pyrometry imaging device, or the like.

[0069] First and second imaging optics **92** and **96** may each include one or more optical devices used to direct light to imaging device **98**. For example, First and second imaging optics **92** and **96** may each include one or more refractive optical device (e.g., a lens), one or more reflective optical device (e.g., a mirror), one or more diffractive optical devices (e.g., a grating), one or more dichroic optical devices (e.g., a dichroic filter or mirror), or the like. Although two sets of imaging optics **92** and **98** are shown in FIG. **2**, in other examples, system **80** may include a single set of imaging optics or more than two sets of imaging optics.

[0070] Occulting device **94** is positioned within the optical train between first imaging optics **92** and second imaging optics **96**. In other example, occulting device **94** may be positioned between imaging device **98** and imaging optics **96** or after before imaging optics **92**. In some examples, occulting device **94** is positioned as the optical component nearest imaging device. This effectively results in removal of the portion of the image which occulting device **94** blocks. In other examples, occulting device **94** is positioned at another position within the optical train **80** where the image of component **22** resolves. Imaging optics **96** then may be configured to image occulting device **94** onto imaging device **98**.

[0071] As shown in FIG. **6**, in some examples, at least a portion of optical system **80** is coaxial with the axis at which energy delivery device **16** outputs energy **34**. For example, at least a portion of second imaging optics **92** (e.g., the portion at which emitted light **104** is incident upon second imaging optics **92**) may be coaxial with the axis at which energy delivery device **16** outputs energy **34**. This may reduce image manipulation that otherwise may be applied to the resulting image to correct for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like. In other examples, optical system **80** (e.g., the portion at which emitted light **104** is incident upon second imaging optics **92**) may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** (FIG. **1**) or another computing device may be configured to manipulate the resulting image to compensate for geometry of component **22**, angular offset of optical system **80** relative to energy delivery device **16**, shadowing due to the angular offset, interference, or the like.

[0072] FIG. **6** also illustrates energy delivery device **16** outputting energy **34**, which is incident upon component **22** and results in formation of melt pool **32**. Surrounding melt pool is a cooling zone **102**, in which temperature gradients from the temperature of melt pool **32** to ambient temperature are present. As shown in FIG. **6**, melt pool **32** and cooling zone **102** emit thermal emissions **104** (e.g., thermal radiation), which travel through optical system **80** to imaging device **98**, which images the thermal emissions **104**. Occulting device **94** occults (e.g., reduces the intensity of or substantially eliminates) thermal emissions **104** from a selected region, e.g., a region corresponding to energy **34** and at least a portion of melt pool **32**. This may allow imaging device **98** to more effectively image relatively lower intensity thermal emissions from at or near the edge of melt pool **32** and within cooling zone **102**. This may enable more accurate measurement of temperatures within the cooling zone **102**, and heat flow within cooling zone **102**.

[0073] Returning to FIG. **5**, system **10** also includes melt pool monitor (“MP monitor”) **82**. Melt pool monitor **82** may include a sensor for monitoring a characteristic of melt pool **32**. The sensor may include an imaging system, such as a visual or thermal camera, e.g., camera to visible light or infrared (IR) radiation. A visible light camera may monitor the geometry of the melt pool, e.g., a width, diameter, shape, or the like. A thermal (or IR) camera may be used to detect the size, temperature, or both of the melt pool. In some examples, a thermal camera may be used to detect the temperature of the melt pool at multiple positions within the melt pool, such as a leading edge, a center, and a trailing edge of the melt pool. In some examples, the imaging system may include a relatively high speed camera capable of capturing image data at a rate of tens or hundreds of frames per second or more, which may facilitate real-time detection of the characteristic of the melt pool.

[0074] FIG. **7** is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique. The technique of FIG. **7** may be implemented by system **10** of FIGS. **1** and **5** and will be described with concurrent reference to FIGS. **1** and **5**. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. **7** may be performed by other systems.

[0075] One or more computing devices **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. **7**). For instance, one or more computing devices **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**. Such parameters may be referred to as powder delivery device operating parameters.

[0076] One or more computing devices **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. Data received from powder flow mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Data from topology sensor **48** indicates powder mass captured by melt pool **32** and added to component **22**.

[0077] One or more computing devices **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from topology sensor, into the mass flow determined based on data received from PFMS **18**.

[0078] Further, one or more computing devices **12** may determine an overall mass flux using the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. **7**).

[0079] Similarly, one or more computing devices **12** may be configured to control energy delivery device **16** to deliver energy **34** to first layer **24** to establish a given heat input (see bottom left of FIG. **7**). For example, one or more computing device **12** may control one or more optical parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**). The one or more optical parameters and the one or more positional parameters related to energy delivery device may be collectively referred to as one or more energy delivery device operating parameters. As such, the one or more energy delivery device operating parameters may include one or more of intensity, pulse rate, pulse width, dwell time at a location, a movement rate relative to first layer **24**, an overlap between adjacent passes of energy **34** across first layer **24**, a pause time between adjacent passes of energy **34** across first layer **24**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**).

[0080] One or more computing devices **12** may be configured to receive from one or more heat sensors, such as optical system **80** and/or melt pool monitor **82**. One or more computing devices may determine a cooling rate and associated heat from using data from optical system **80** and may determine a heat input into component using a size and/or temperature of melt pool **32** as observed by melt pool monitor. One or more computing devices **12** may be configured to determine an overall heat flux using these data. One or more computing devices **12** then may use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see top left of FIG. **7**).

[0081] In some examples, one or more computing devices **12** also may use the deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the heat flux, as the added powder mass and quench effects associated with the captured powder affect the cooling rate.

[0082] FIG. **8** is a flow diagram illustrating an example technique for closed loop control of an additive manufacturing system, in accordance with one or more aspects of this disclosure. The technique of FIG. **8** may be performed by one or more processors of a computing device, such as computing device **12** of FIG. **1** of FIG. **5**.

[0083] Computing device **12** may control, based on a plurality of operating parameters, an energy delivery device to delivery energy to a build surface of a component to form a melt pool (**802**). For instance, computing device **12** may control energy delivery device **16** to operate in accordance with one or more energy delivery device operating parameters. As discussed above, the energy delivery device operating parameters may include one or more of an intensity, a pulse rate, a pulse width, a dwell time at a location, a movement rate, an overlap between adjacent passes, and/or a pause time between adjacent passes.

[0084] Computing device **12** may control, based on the plurality of operating parameters, a powder delivery device to direct a powder stream toward the melt pool (**804**). For instance, computing device **12** may control powder delivery device **14** to operate in accordance with one or more powder delivery device operating parameters. As discussed above, the powder delivery device operating parameters may include one or more of a speed of an agitator of a powder source, a gas flow rate of gas flowing through the powder source, a position of one or more valves within a flow path, or the like to control a powder feed rate output by the powder source.

[0085] Computing device **12** may determine a mass flux (**806**). For instance, as discussed above with reference to FIG. **7**, computing device **12** may determine the mass flux using the data received from PFMS **18**, powder flow mass sensor **44**, and/or topology sensor **48**.

[0086] Computing device **12** may determine a heat flux (**808**). For instance, as discussed above with reference to FIG. **7**, computing device **12** may determine the heat flux using the data received from the one or more heat sensors, such as optical system **80** and/or melt pool monitor **82**.

[0087] Computing device **12** may input, into one or more machine learning models, the mass flux and the heat flux (**810**). For instance, computing device **12** may input the mass flux and/or the heat flux into an input layer (e.g., input layer **202**) of ML model **200**. In some examples, computing device **12** may, in addition to or in place of the mass flux and/or heat flux, input the measurements from which the mass and heat flux were calculated into the input layer of ML model **200**. As such, in some examples, computing device **12** may omit intermediate calculation of the mass and/or heat flux.

[0088] In some examples, the machine learning model(s) used (e.g., ML model **200**) may be generic ML models for additive manufacturing. However, when manufacturing certain complex components (e.g., members of gas-turbine engines), such generic models may not produce optimal results. In accordance with one or more aspects of this disclosure, in some examples, computing device **12** may use machine learning models that are specific to the type of component being built. For instance, ML model **200** may be trained on components of a same type as the component being built.

[0089] Computing device **12** may update, based on output from the one or more machine learning models, two or more of the plurality of operating parameters (**812**). For instance, computing device **12** may update two of more of the plurality of operating parameters based on an output (e.g., output **207**) of ML model **200**. In some examples, the output of ML model **200** may include an output for each of the operating parameters controlled by computing device **12**. With each iteration of this technique, some of the operating parameters may be changed (e.g., have new values) while some of the operating parameters may not be changed (e.g., have the same values). Computing device **12** may then control the energy delivery device (**802**) and the powder delivery device (**804**) based on the updated operating parameters.

[0090] As discussed above, in some examples, the two or more operating parameters updated by computing device **12**, in parallel, may have a non-linear relationship. For instance, adjusting the delivered energy may have a non-linear relationship to either powder delivery feed rate or to travel speeds needed to compensate for the change in delivered energy (e.g., to control the melt pool size or powder capture rate). As discussed above, adjusting multiple parameters, especially non-linear parameters, in parallel may be taught away from as undesirable. However, by utilizing one or more machine learning models as described herein, multiple parameters may be successfully adjusted in parallel.

[0091] Computing device **12** may perform the operating parameter update at regular intervals. As one example, computing device **12** may perform the operating parameter update at temporal intervals (e.g., every 5 seconds, 10 seconds, 1 minutes, 5 minutes, etc.). As another example, computing device **12** may perform the operating parameter update at build intervals (e.g., every layer, every second layer, etc.). By adjusting two or more operating parameters in parallel, the controller may more quickly “react” to changes in the build.

[0092] As explained above, additive manufacturing systems described herein may be configured to use machine learning processes for evaluating sensor data to adjust operating parameters for subsequent deposition. FIG. **9**A is a conceptual diagram illustrating an example machine learning model according to one or more aspects of this disclosure. Machine learning model **200** may be an example of a machine learning model implemented by computing device **12**. Machine learning model **200** may be an example of a deep learning model, or deep learning algorithm, trained to determine deposition parameters for subsequent layers. Computing device **12** and/or another device, may train, store, and/or utilize machine learning model **200**, but other devices of system **10** may apply inputs to machine learning model **200**. In some examples, other types of machine learning and deep learning models or algorithms may be utilized. For example, a convolutional neural network model of ResNet-18 may be used. Some non-limiting examples of models that may be used for transfer learning include AlexNet, VGGNet, GoogleNet, ResNet50, or DenseNet, etc. Some non-limiting examples of machine learning techniques include Support Vector Machines, K-Nearest Neighbor algorithm, and Multi-layer Perceptron.

[0093] As shown in the example of FIG. **9**A, machine learning model **200** may include three types of layers. These three types of layers include input layer **202**, hidden layers **204**, and output layer **206**. Output layer **206** includes the output from the transfer function **205** of output layer **206**. Input layer **202** represents each of the input values X**1** through X**4** provided to machine learning model **200**. In some examples, the input values may include any of the values input into the machine learning model, as described above. For example, the input values may include sensor data from sensor **48**, or other data received by computing device **12**, as described above. In some examples, the input values may be determined heat flux, determined mass flux, or values used to determine the mass flux or the heat flux. In addition, in some examples input values of machine learning model **200** may include additional data, such as other data that may be collected by or stored in system **10**.

[0094] Each of the input values for each node in the input layer **202** is provided to each node of a first layer of hidden layers **204**. In the example of FIG. **5**A, hidden layers **204** include two layers, one layer having four nodes and the other layer having three nodes, but fewer or greater number of nodes may be used in other examples. Each input from input layer **202** is multiplied by a weight and then summed at each node of hidden layers **204**. During training of machine learning model **200**, the weights for each input are adjusted to establish a relationship between topological data and deposition parameters that generate the layer represented by the topological data. In some examples, one hidden layer may be incorporated into machine learning model **200**, or three or more hidden layers may be incorporated into machine learning model **200**, where each layer includes the same or different number of nodes.

[0095] The result of each node within hidden layers **204** is applied to the transfer function of output layer **206**. The transfer function may be linear or non-linear, depending on the number of layers within machine learning model **200**. Example non-linear transfer functions may be a sigmoid function or a rectifier function. The output **207** of the transfer function may be a set of updated operating parameters.

[0096] FIG. **9**B is a conceptual diagram illustrating an example training process for a machine learning model according to one or more aspects of this disclosure. Process **210** may be used to train machine learning model **200**. Machine learning model **200** may be implemented using any number of models for supervised and/or reinforcement learning, such as but not limited to, an artificial neural network, a decision tree, naïve Bayes network, support vector machine, or k-nearest neighbor model, CNN, RNN, LSTM, ensemble network, to name only a few examples.

[0097] In some examples, computing device **12** or another device trains machine learning model **100** based on a corpus of training data **212**. Training data **212** may include, for example, previous topological data of a build over time, previous operating parameter data, and/or the like. Previous operating parameter data, for example, may include operating parameters associated with previous additive manufacturing processes. In some examples, the sensor data and/or operating parameter data may be generated by executing a natural language processing application on content of a plurality of operational records to automatically extract relevant nor desired training data. For example, by using an NLP model, computing device **12** may capture potentially cofounding operation conditions or factors, which may be useful in determining a set of operating parameters. Computing device **12** may use the NLP model to capture such details.

[0098] In some examples, training data **212** may include annotations identifying effects of operating parameters indicated in topological data of training data **212**. Training data **212** may include data from past additive deposition processes performed on components having different geometries, formed from different materials, formed under different conditions, and/or the like.

[0099] While training machine learning model **200**, computing device **12** may compare **214** a prediction or classification with a target output **216**. Computing device **12** may utilize an error signal from the comparison to train (learning/training **218**) machine learning model **200**. Computing device **12** may generate machine learning model weights or other modifications which computing device **12** may use to modify machine learning model **200**. For example, computing device **12** may modify the weights of machine learning model **200** based on the learning/training **218**.

[0100] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0101] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0102] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to

implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.
[0103] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).
[0104] Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. An additive manufacturing system comprising: an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of the component; a powder delivery device configured to direct a powder stream toward the melt pool; a plurality of sensors, each mass sensor associated with a portion of the additive manufacturing system; a plurality of heat sensors; and one or more computing devices configured to: receive data from the plurality of mass sensors; determine an overall mass flux based on the data from the plurality of mass sensors; receive data from the plurality of heat sensors; determine an overall heat flux based on the data from the plurality of heat sensors; and input, into one or more machine learning models, the overall mass flux and the overall heat flux; and control, based at least in part on outputs from the one or more machine learning models, the powder delivery device and the energy delivery device.
2. The additive manufacturing system of claim 1, wherein, to control the powder delivery device and the energy delivery device, the one or more computing devices are configured to: control a plurality of operating parameters of the powder delivery device and the energy delivery device.
3. The additive manufacturing system of claim 2, wherein, to control the plurality of operating parameters, the one or more computing devices are configured to: adjust, in parallel, two or more operating parameters of the plurality of operating parameters.
4. The additive manufacturing system of claim 3, wherein the two or more operating parameters have a non-linear impact on building of the component.
5. The additive manufacturing system of claim 2, wherein the plurality of operating parameters includes one or more powder delivery device operating parameters and one or more energy delivery device operating parameters.
6. The additive manufacturing system of claim 5, wherein: the one or more powder delivery device operating parameters include one or more of a powder feed rate, a gas flow rate, and an agitator rate, and the one or more energy delivery device operating parameters include one or more of power, travel speed, pause time, dwell time, working distance, and spot size.
7. The additive manufacturing system of claim 1, wherein the one or more computing devices are further configured to update one or both of a layer thickness and build strategy of the component based at least in part on the outputs from the one or more machine learning models.
8. The additive manufacturing system of claim 1, wherein the one or more machine learning models are trained on components of a same type as the component.
9. The additive manufacturing system of claim 8, wherein the one or more computing devices are further configured to: update, based on the build of the component, the one or more machine

learning models.
10. The additive manufacturing system of claim 8, wherein the component is a member of a gas-turbine engine.
11. A method comprising: receiving, by one or more computing devices, data from a plurality of mass sensors of an additive manufacturing system, wherein the additive manufacturing system comprises an energy delivery device configured to deliver energy to a build surface of a component to form a melt pool in the build surface of a component, a powder delivery device configured to direct a powder stream toward the melt pool, the plurality of mass sensors, each mass sensor associated with a portion of the additive manufacturing system, and a plurality of heat sensors; determining, by the one or more computing devices, a mass flux based on the data from the plurality of mass sensors; receiving, by the one or more computing devices, data from the plurality of heat sensors; determining, by the one or more computing devices, a heat flux based on the data from the plurality of heat sensors; inputting, by the one or more computing devices and into one or more machine learning models, the mass flux and the heat flux; and controlling, based at least in part on outputs from the one or more machine learning models, the powder delivery device and the energy delivery device.
12. The method of claim 11, wherein controlling the powder delivery device and the energy delivery device comprises: controlling a plurality of operating parameters of the powder delivery device and the energy delivery device.
13. The method of claim 12, wherein controlling the plurality of operating parameters comprises: adjusting, in parallel, two or more operating parameters of the plurality of operating parameters.
14. The method of claim 13, wherein the two or more operating parameters have a non-linear impact on building of the component.
15. The method of claim 12, wherein the plurality of operating parameters includes one or more powder delivery device operating parameters and one or more energy delivery device operating parameters.
16. The method of claim 15, wherein: the one or more powder delivery device operating parameters include one or more of a powder feed rate, a gas flow rate, and an agitator rate, and the one or more energy delivery device operating parameters include one or more of power, travel speed, pause time, dwell time, working distance, and spot size.
17. The method of claim 11, further comprising updating one or both of a layer thickness and build strategy of the component based at least in part on the outputs from the one or more machine learning models.
18. The method of claim 11, wherein the one or more machine learning models are trained on components of a same type as the component.
19. The method of claim 18, further comprising: updating, based on the build of the component, the one or more machine learning models.
20. The method of claim 18, wherein the component is a member of a gas-turbine engine.